What the system is, and for whom

HyperspaceVault · Architecture · View 1 of 3 — Conceptual

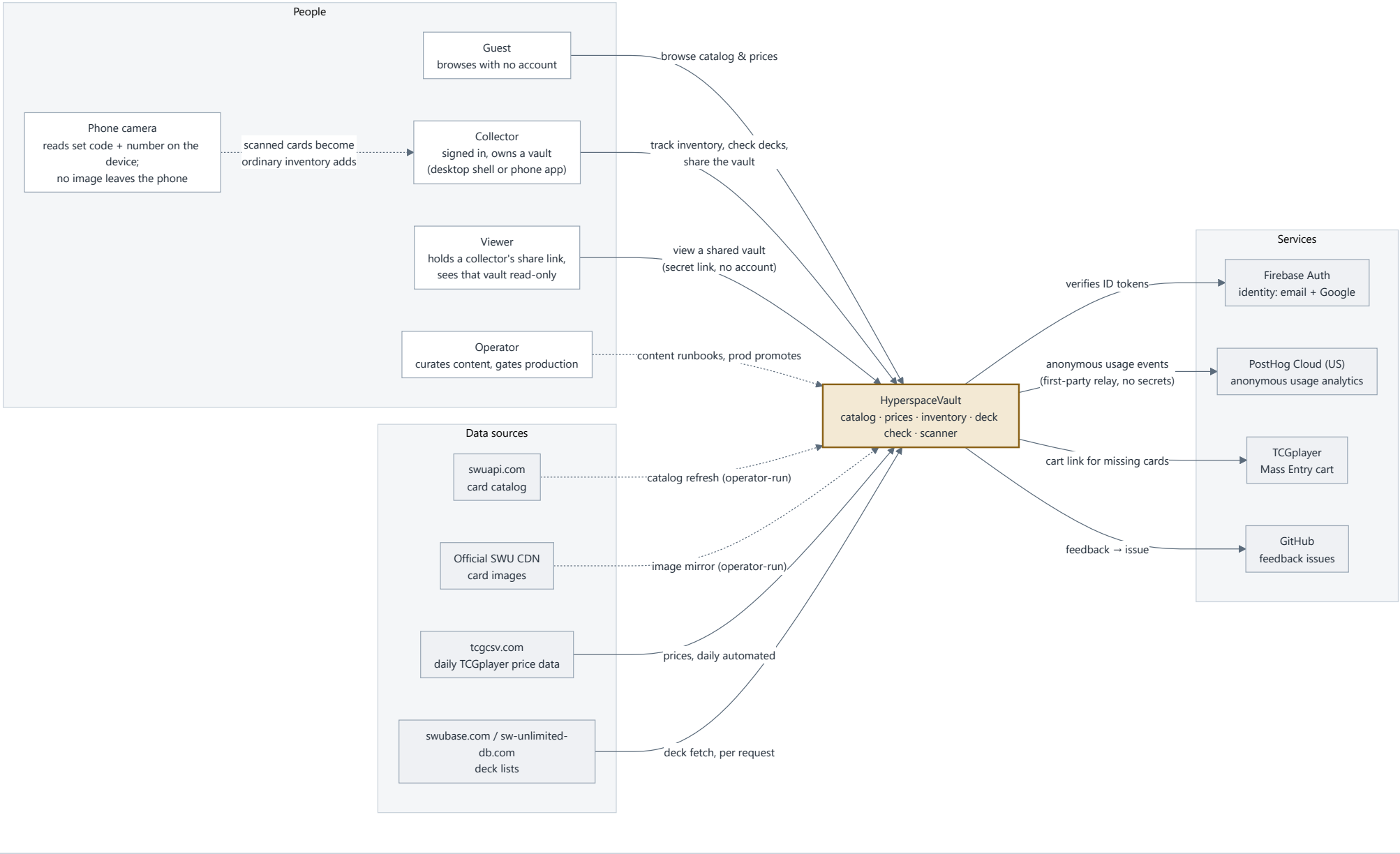
HyperspaceVault is a multi-tenant web application for Star Wars: Unlimited collectors. It carries the complete card catalog with daily market prices, and gives each signed-in collector a private, isolated vault to track what they own, what a deck would cost them, and what they're missing — on a desktop browser or, since v2.0, as a phone app with a card scanner.

Prepared 2026-07-28 · Updated 2026-10-18 — as of release v2.0.1 (prod `c65cf87`) with the v2.1 candidate on dev (main `d3118c5`) · Companion views: Logical (components & data) · Physical (deployment & delivery)

System context

Three kinds of people touch the system, and it leans on a small set of outside services — one for identity, several as data sources, one for anonymous usage analytics, and one as an outbound hand-off when a collector wants to actually buy the cards they're missing.

Context — actors and external systems


People
Guest
browses with no account
Phone camera
reads set code + number on the device;
no image leaves the phone
scanned cards become ordinary inventory adds
Collector
signed in, owns a vault
(desktop shell or phone app)
Viewer
holds a collector's share link, sees that vault read-only
Operator
curates content, gates production
browse catalog & prices
track inventory, check decks, share the vault
view a shared vault
(secret link, no account)
content runbooks, prod promotes
Data sources
swuapi.com
card catalog
Official SWU CDN
card images
tcgcsv.com
daily TCGplayer price data
swubase.com / sw-unlimited-db.com
deck lists
catalog refresh (operator-run)
image mirror (operator-run)
prices, daily automated
deck fetch, per request
HyperspaceVault
catalog · prices · inventory · deck check · scanner
verifies ID tokens
anonymous usage events
(first-party relay, no secrets)
cart link for missing cards
feedback → issue
Services
Firebase Auth
identity: email + Google
PostHog Cloud (US)
anonymous usage analytics
TCGplayer
Mass Entry cart
GitHub
feedback issues

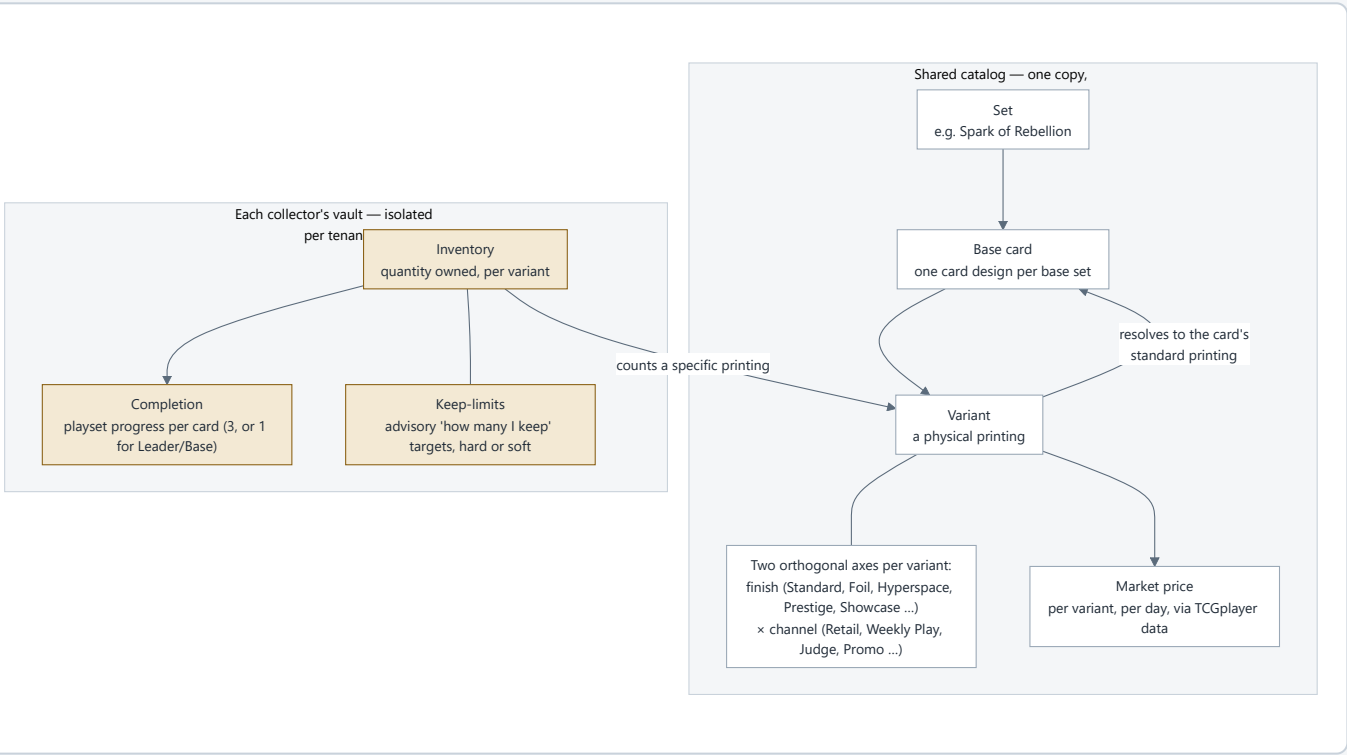
Dashed lines are **operator-gated** flows — the card catalog and images only change when a human runs the content runbook. Solid inbound lines are automated or per-request. That split is deliberate: market prices move daily on their own, but the catalog is treated as curated content.

One application, two layouts. The same deployment serves a desktop shell and a phone app. A viewport narrower than 768 px, or a touch-first device whose screen's short side is 500 px or less, gets the phone app — its own navigation, a keypad for fast adds, and the camera scanner; everything else gets the desktop shell. The phone camera is the one piece of the picture that is a *device*, not a service: card recognition runs on the phone itself, and what reaches the system is an ordinary inventory add.

The domain in one picture

Everything in the system divides into two worlds: a **shared catalog** — one copy, visible to everyone including guests — and **per-collector vault data**, isolated per tenant. The catalog's central idea is that one card design (a *base card*) exists as many physical *printings* (variants), and every variant resolves back to its card's standard printing so quantities and prices can roll up.

Domain concept map



Card vs. variant vs. finish A base card is the design; variants are its printings. A variant's identity is described by two independent axes — visual finish and distribution channel — derived from the source data by one shared classifier, never stored redundantly.

Standard-variant mapping Every printing walks its "variant of" chain to an ultimate root (chains up to two hops exist in the wild), landing on the card's Standard printing. The one genuine exception in the whole catalog is flagged, not blocked.

Pricing, two ways *Standard* shows the standard printing's market price; *cheapest* shows the least money out the door — the lowest listed price across all of a card's printings. Every priced surface carries TCGplayer attribution and an as-of date; unpriced cards are excluded from sums with a visible count, never silently zeroed.

Completion vs. keep-limits Deliberately decoupled: completion is variant-agnostic playset progress; keep-limits are each collector's own "how many do I keep" policy, enforceable as a hard block or a soft over-limit flag.

Deck check Paste a deck (or a supported deck-site URL) and get a diff against your vault — what's missing, what it costs, and a pre-filled TCGplayer cart. Ephemeral by design: nothing is persisted.

Scanning (phone) Point the camera at a card; the phone reads the printed set code and card number itself and offers the match for a one-tap add. Recognition is entirely on-device (the reader and its models ship with the app and are fetched only when the camera is first used); the system never sees an image.

Who can do what

The catalog is an open storefront window — browsing requires nothing. Progressive trust gates the rest: an account, then a verified email for anything that mutates a vault, then a fresh re-authentication for the one irreversible action.

Capability	Requires	Notes
Browse catalog, prices, images	`GUEST`	Public and CDN-cached; the growth funnel
View a shared vault	`SHARE LINK`	Secret link is the whole credential — read-only, revocable, rate-limited (v1.4)
Send feedback	`GUEST`	Rate-limited; lands as a GitHub issue
See own quantities on cards, deck check	`SIGNED IN`	First sign-in auto-provisions the vault
Add / remove cards, import / export	`SIGNED IN + VERIFIED EMAIL`	All vault mutations sit behind verification
Scan cards with the phone camera	`SIGNED IN + VERIFIED EMAIL`	Phone app only (v2.0); recognition on-device, nothing uploaded; each confirmed card is an ordinary verified-email inventory add
Keep-limit settings	`SIGNED IN`	Per-tenant policy, hard or soft mode
Share the vault (view-only link)	`SIGNED IN`	One active link; rename, rotate, or revoke at will
Delete account	`SIGNED IN + RECENT RE-AUTH`	5-minute freshness window; full tenant purge
Catalog refresh, image mirror, prod release	`OPERATOR`	Runbook-driven; production is always a human gate unless a release is explicitly low-risk

Principles that shape everything downstream

- **Catalog is curated, prices are automated.** Card data changes only through an operator-run runbook with verification steps; market prices refresh themselves daily.
- **Isolation lives in the database, not the application.** Tenant separation is enforced by Postgres row-level security — application bugs cannot leak one vault into another.
- **Guests are first-class.** The entire catalog experience works with no account, which sets the caching, cost, and abuse-control posture of the whole platform — and is why the analytics relay accepts events from

logged-out visitors.

- **Honest numbers.** Prices always attributed and dated; missing data shown as missing, never as zero.
- **The phone is a full client, not a cut-down one.** The phone app has its own navigation, a fast-add keypad and the scanner; recognition stays on the device, so the privacy posture of the catalog ("we never see your cards") holds for the camera too.
- **Telemetry is anonymous and never load-bearing.** Usage events go through a first-party relay that holds no secrets; if analytics fails, the app does not notice.
- **Production is a decision.** Every change reaches the dev environment automatically; reaching collectors requires an explicit owner action — or a release where every commit was pre-declared low-risk.

> **Where to go deeper:** the Logical view maps these concepts onto components, APIs, and the data model; the Physical view shows the cloud footprint and how a change travels from pull request to production.

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